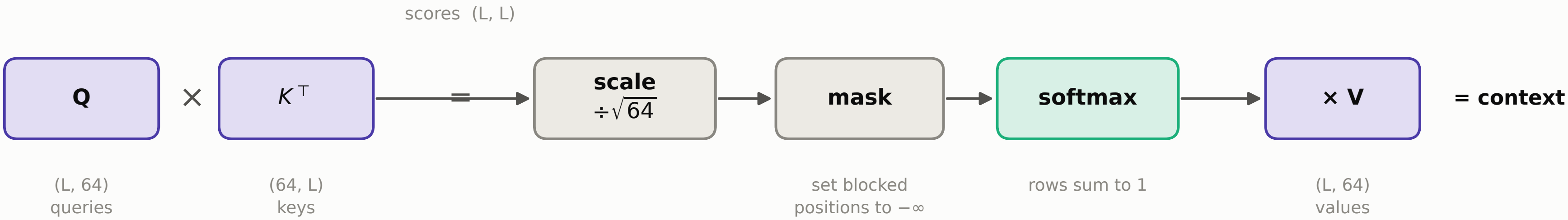
Scaled dot-product attention

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$



Why divide by $\sqrt{d_k}$?

A dot product of two 64-dimensional unit-variance vectors has variance 64, so raw scores swing over roughly ±8. Softmax of values that spread out is almost one-hot, and its gradient p(1−p) collapses to zero — the layer stops learning. Dividing by √64 = 8 restores unit variance and keeps softmax in its responsive range.

Read it as a soft dictionary lookup

Each query is compared against every key by dot product. A hard lookup would take the single best-matching key; softmax takes a weighted blend instead — which is what makes the whole operation differentiable, and therefore trainable end to end.